LEAF: A Robust Multimodal Federated Learning Framework for Tea Pest and Disease Diagnosis

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Abstract—**Tea pest and disease diagnosis draws on both lesion imagery and written field notes, but small agricultural corpora make multimodal and privacy claims easy to overstate. We present LEAF (Leakage-audited Evidence-Aware Fusion), a reliability-routed multimodal federated framework for tea pest and disease diagnosis, and report a leakage-audited comparison of five vision–language systems on one five-class tea corpus, under source-grouped splitting, exact image-box/text linkage, missing-modality tests, reliability-aware fusion, and measured FedAvg adaptation. The corpus holds 371 crops from 200 photographs, split 222/74/75 across 122/40/38 disjoint sources; every record links to its exact box, and a training-fitted filter masks 45 target-shortcut tokens; 61% of note sentences occur in more than one class, so the notes inform the label without determining it. On the identical 75-crop support, frozen DistilBERT, frozen ViT-tiny, the raw ResNet–Transformer VLM, a frozen ViT-tiny–DistilBERT VLM, and the routed ResNet–Transformer reach 24.00%, 64.00%, 58.67%, 64.00%, and 61.33% accuracy. Selection is validation-only: the ViT–DistilBERT candidate's 72.97% validation accuracy stays below the routed incumbent's 77.03%, so the incumbent is retained; on test the candidate leads by 2 crops of 75, an 11–9 discordance that is not significant (McNemar $p = 0.82$). Fusion of frozen encoders is what pays: on that support ViT-Base with BERT-small reaches 80.00% against 73.33% image-only and 64.00% text-only, beating both parents outright rather than only under note sparsity. LEAF, the trained router, does not exploit the note channel the same way — mismatching the note costs it only 2.67 points — though ablating either modality moves predictions (41 of 75 for the note, 49 for the image). LEAF retains 84–94% of its 0.6919 centralized validation macro F1 under FedAvg over 2–8 simulated clients. Across five federated architectures, federation beats local-only training at 18.9–67.2 MiB per client-round, and no aggregator dominates. Under severe skew, client dropout, straggler staleness and poisoned updates, fusion absorbs the first two and degrades under the last two. However, none of the tested configurations demonstrated reliable tolerance to Byzantine clients. Nothing saturates, but clients are simulated and notes are templated, so multi-site validation remains necessary.**

Index Terms—**Federated Learning, Vision-Language Models, Tea Pest and Disease Classification, Multimodal Learning, Evaluation Protocol, Data Leakage, Precision Agriculture**

I. Introduction

TEA (*Camellia sinensis*) is a vital cash crop for millions of smallholder farmers across South Asia, East Africa, and East Asia. Five visually confusable stress classes — leaf blight, leaf hoppers, leaf rust, looper caterpillars, and mosquito bug — reduce yield and leaf quality across major growing regions [1], [2]. Recent diagnostic systems improve field deployment through lightweight CNNs, attention modules, and edge-oriented pipelines [1]–[3], [5], [6], [8], but nearly all begin and end with the photograph, whereas staff record lesion shape, flush condition, weather history and pest pressure alongside it [50], [51]. Centralizing every image and note is also unattractive for estates with limited connectivity and proprietary records [52], [53]; Federated Learning [15] addresses that, but FL work on tea pathology has not integrated the two kinds of evidence.

A second, methodological motivation dominates. One source photograph may contain several annotated regions, so crop-level random splits place near-identical leaf context on both sides of the boundary. Text can leak the target even more directly when disease names or class-pure terms remain in descriptions. We therefore group by source image, link text at the exact box level, and learn the blocked vocabulary from training annotations only.

Contributions. We advance LEAF, a multimodal federated framework whose components are selected by the comparison reported here rather than asserted. We provide (i) exact crop–note linkage with source-disjoint splits and train-only leakage control; (ii) common-support unimodal, paired, mismatched, and model-family comparisons, extended by missing-modality and graded-corruption probes that test whether either branch is used at all; (iii) explicit visual and sparse-note routing; and (iv) three-seed client scaling plus a matched evaluation suite covering partitioning, anti-collapse components, warm start, fusion, communication, and four FL aggregators against local-only.

II. Related Work

Image recognition. Tea and plant disease work has moved from backbone benchmarking [3], [4] toward efficiency and deployment: attention-guided lightweight detectors [1], [45], [48], ensembles [6], [7], transformer hybrids [46], [49], real-time field datasets [5], IoT-assisted management [8], edge distillation and quantization [38], [40], and domain adaptation [39], [41]. Surveys map the breadth [2], [9]; across it the work is image-only, centralized, and split at crop or patch level, so one photograph can place near-identical regions on both sides of the boundary.

Agricultural federation. FedAvg [15] is extended with explainability [10], hierarchical sensing [11], blockchain-backed updates [12], cloud coordination [13], and coordinator-free edge federation [14], with trust and continual-update duties reviewed in [42], [47]. Those clients are almost always image-only, and federation is reported as headline accuracy rather than measured cost: communication per round, warm start, realized skew and matched-budget aggregator comparison are rarely reported together. We report them here.

Vision–language fusion. Contrastive alignment [22], query transformers over frozen encoders [23], gated cross-attention [24], joint captioning [25], and unified interfaces [26] supply the fusion vocabulary, realized here through compact encoders [17], [19], [29]–[33]. Agricultural instances include multimodal LLM fine-tuning [27], federated feature replay over frozen CLIP [28], sensor–image fusion [43], and knowledge-driven semantic classification [44]. Their recurring weakness is the text side: notes generated from the class label or paired by class matching are partial copies of the target. Ours is templated too, which is why pairing is fixed at the exact box index and the shortcut vocabulary fitted on training text alone. The gap is paired image–text FL with a source-disjoint leakage audit.

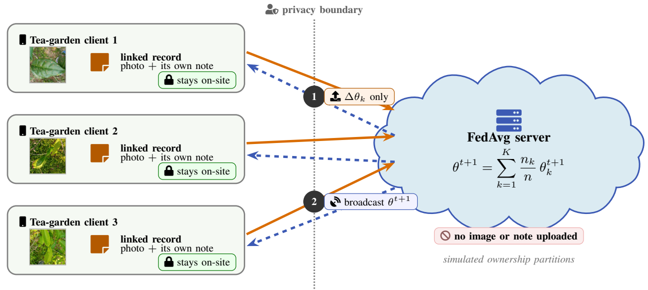

Fig. 1: Federated deployment workflow using representative corpus photographs. Each simulated tea-garden client keeps linked photographs and field notes on-site; only model updates $\Delta\theta_k$ cross the explicit privacy boundary, and the FedAvg server returns θ^{t+1}.

TABLE I: Source-grouped paired-data audit

Split	Sources	Crops	Blight	Hoppers	Rust	Looper/Mosq.
Train	122	222	77	5	40	61/39
Validation	40	74	26	2	14	20/12
Test	38	75	26	2	13	21/13
Total	200	371	129	9	67	102/64

III. System Model

Each estate keeps images and notes local; only model updates cross the boundary (Fig. 6). The controlled ablations use one selected checkpoint, while FL adapts its trainable multimodal states with the visual backbone frozen. BLAKE2b maps field-note words to 30 522 IDs in 128-token sequences; ImageNet-normalized 224×224 inputs yield 49 ResNet-50 spatial tokens for bidirectional cross-attention.

A. Dataset and Leakage Audit

Held fixed. Every table except Table II is scored on the same 222/74/75 source-grouped crops, the same exact box–note pairing and the same train-fitted leakage mask, so any difference between them is the axis that table varies. The matched-setting federated suite of Table II re-partitions those same 371 observations as 260/55/56 over 199 source groups, and is read only against itself.

The audited corpus (Table I) contains 200 field photographs collected from the IIT Kharagpur tea garden and annotated with YOLO oriented bounding boxes. One crop is extracted per valid box with 10% context padding, giving 371 examples across *LEAF_BLIGHT* (129), *LEAF_HOPPERS* (9), *LEAF_RUST* (67), *LOOPER_CATERPILLARS* (102), and *MOSQUITO_BUG* (64). Classes are the raw YOLO identifiers under the supplied annotation schema; the *LEAF_* prefix belongs to those identifiers and is unrelated to LEAF.

Seed 42 splits source filenames, never crops: overlap is zero and the test holds 75 crops from 38 unseen photographs. All 371 records link by exact (photograph, box index); the nine-example leaf-hopper class remains the bottleneck. Notes are templated from manually written phrase pools, not independent farmer prose. Recording observations against a controlled vocabulary is standard practice: crop trait ontologies supply harmonized terms [54], [55] and severity is scored on fixed ordinal scales [56], [57]. Templating is therefore a fair proxy for the *structure* of real records, not their diversity: the 371 notes draw on 140 distinct sentences, 61% of which occur in more than one class, and a class's diagnostic sign appears in only 45% of its notes. A bag-of-words classifier consequently reaches 0.640 rather than saturating, so the note channel is informative without being label-determining, and the comparison between modalities is not decided by the corpus. A 178-token target-pure vocabulary is learned on training text only and then frozen. No external tea images are pooled; ImageNet supplies weights only.

B. Model Variants and Objective

The reliability network is a two-layer MLP over the concatenated embeddings whose two softmax-normalized outputs give weights $r_t + r_v = 1$ — how far the fused prediction leans on the note and on the image. An absent modality's logit is set to -10^4 before the softmax, so its weight goes to zero and the survivor takes the full unit of mass. The final logits are

$$\mathbf{z} = \mathbf{z}_f + \lambda\left(r_t \mathbf{z}_t + r_v \mathbf{z}_v\right), \qquad \lambda = 2, \tag{1}$$

where $\mathbf{z}_f$ is the fusion head's five-way output and $\mathbf{z}_t$, $\mathbf{z}_v$ come from two linear *expert* heads reading each embedding directly. The residual weight $\lambda = 2$ is fixed rather than learned, so the two single-modality experts count twice the fused head and a usable prediction survives when one branch is absent; a missing modality's expert is zeroed and the residual stays active at test time.

Comparison models. Frozen DistilBERT [33] and ViT-tiny [17] use validation-selected ridge heads. Their late-fusion candidate uses train-fitted 64-component projections and one of 360 validation-screened linear heads; test embeddings are extracted only after the choice is frozen. The raw and routed VLM rows are the cross-attention output before and after the reliability router.

For one-hot label $\mathbf{y}$, training uses class-weighted, label-smoothed cross entropy — weights inverse to class frequency, smoothing 0.1 — with auxiliary parent losses and supervised cross-modal alignment:

$$\mathcal{L} = \mathcal{L}_f + 0.20\mathcal{L}_t + 1.25\mathcal{L}_v + 0.05\mathcal{L}_{align}. \tag{2}$$

Here $\mathcal{L}_f$ is the loss on the fused logits; $\mathcal{L}_t$ and $\mathcal{L}_v$ apply it to the two expert heads, averaged only over samples where that modality is present, so a text-only batch never trains the image expert; $\mathcal{L}_{align}$ is a supervised contrastive term over the ℓ_2-normalized paired embeddings, evaluated only when a batch holds at least two paired crops. The three weights are fixed by construction rather than tuned: image-only learning is the harder path while templated notes make the text expert easy to satisfy, and alignment is a regularizer that must not compete with the classification objective. Modality dropout samples image-only, text-only and paired batches with probabilities 0.50, 0.10, 0.40; checkpoints are selected by $0.65F_{1,\text{paired}} + 0.35F_{1,\text{image}}$ on validation.

For K clients, the same local objective is aggregated with FedAvg [15],

$$\theta^{t+1} = \sum_{k=1}^{K} \frac{n_k}{n} \theta_k^{t+1}, \tag{3}$$

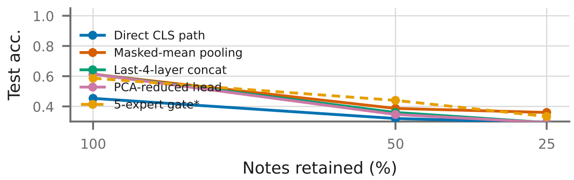

Fig. 2: Text-only rungs under the same deletion, on the identical 75 crops. Every rung falls from the ceiling together: the best single rung drops 0.613 to 0.360 as three quarters of the tokens go. Gates are dashed and marked $*$.

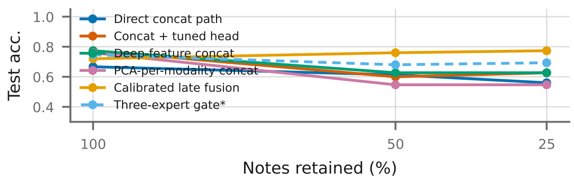

Fig. 3: Text+image rungs under the same deletion. Fusion degrades far more gently: the best rung holds 0.760 or better at every sparsity where the best text rung falls to 0.360.

where n_k is client k's sample count. FL starts from the selected checkpoint, freezes ResNet-50 and averages the remaining states after one local epoch: adaptation, not training from scratch.

IV. Evaluated Systems

A. Base Cross-Modal Checkpoint

The checkpoint trains in PyTorch [37] for at most 15 AdamW epochs (batch 16, peak 10^{-4}). Two cached augmentations per training crop remain views of 222 crops, not new samples.

B. Enhanced Image-Only Inference

Five single deployable visual systems compete on the common support, ranked by the pre-declared validation score $0.6F_1 + 0.4$ accuracy, with the locked test support reported after the choice is frozen. Across the eight visual rungs, locked test accuracy rises from 0.613 to 0.720 (Fig. 4); Figs. 2 and 3 are the corresponding text and text+image ladders. *Cand.* counts the configurations each rung searched, and the gate rungs route among several encoders, which is why they are marked as diagnostics.

The same ladder runs for text and text+image on the identical support, so every rung is comparable with the others and with the visual ladder. Neither ladder is flat now that the notes no longer determine the label. The best text rung reaches 0.613 at complete notes and 0.360 once three quarters of the tokens are gone, while the best fusion rung holds 0.773 at both (Fig. 3): the fused pipeline is what survives note loss, and the validation-to-test gap still widens with search (Fig. 5).

The calibrated base wins on both sides of the split on the smallest search in the table (validation 0.771, test accuracy 0.680); the gates score higher only by fitting more routings on 74 crops, and Fig. 5 shows what that costs. We carry the calibrated base as the visual path and report the gate as a diagnostic ceiling.

C. Reliability-Aware Sparse-Note Rule

For sparse notes, a stable hash retains each non-special token with probability 0.50. A validation-fixed rule routes low-confidence (< 0.60), base-predicted leaf-rust cases to the sparse-text expert; otherwise it keeps late fusion. Because the routing family was refined on this internal partition, its test score is **exploratory and non-pristine**.

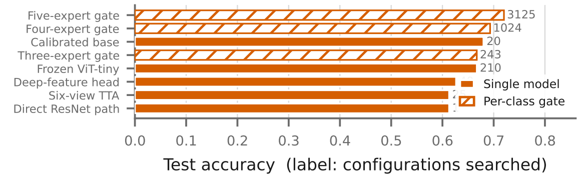

Fig. 4: The image-only ladder. Rungs are alternative methods rather than stages of one progression, so they are compared as bars rather than joined by a line; the search size each rung fitted is annotated. Accuracy rises 0.613 to 0.720 across three decades of search. The ladder carries no note axis because the image branch does not move when notes are deleted. Hatched bars are per-class gates, which are diagnostics rather than deployable models.

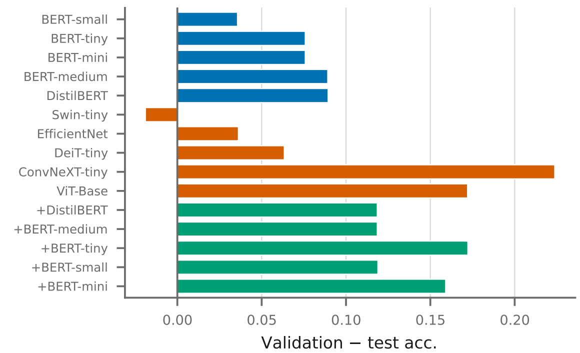

Fig. 5: Validation minus test accuracy per system on the aligned support. A large positive gap is selection that did not transfer from 74 validation crops.

D. Multimodal Federated Adaptation

FL adapts 8.69M parameters with frozen ResNet-50, one local epoch, full participation, and sample weighting. We sweep $K = \{2, 3, 5, 8\}$ for eight rounds over seeds 42, 123, and 2026 on the 74-crop validation set. Clients are simulated; the ensemble and router are excluded from this FL score.

The two halves of a round separate cleanly: every stage touching a photograph, box, or note runs on the client.

E. Advisory Module

The Classify–Retrieve–Advise module embeds a query with Sentence-BERT [35] and performs exact FAISS inner-product search [34], [36]. Its corpus is textual only and shares no records with the box-linked crop set: 673 unique observations (132–138 per class) built from 131 distinct templated sentences. A deduplicated split leaves 472 passages and 201 queries with no shared observation. Correctness requires a class match.

V. Performance Evaluation

A. Protocol

All crops from a source photograph remain in one partition. Model selection and the five-expert visual gate use only the 74-crop validation partition; every number below uses the same 75-crop, 38-source internal test support. Macro F1 accompanies accuracy because the classes are imbalanced.

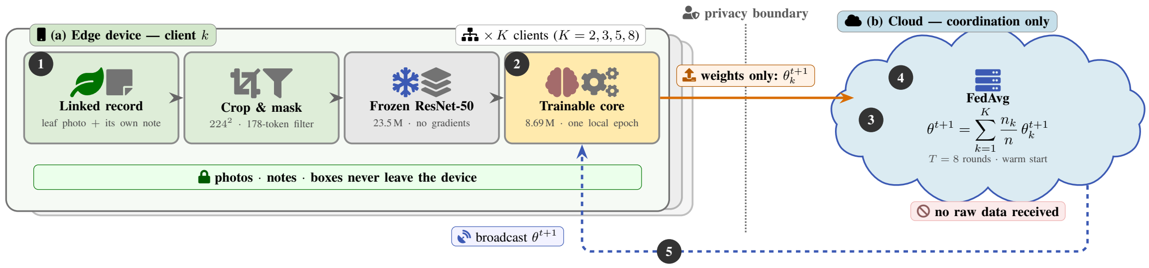

Fig. 6: Division of labour in one FedAvg round. The edge device runs every data-touching stage — box-linked loading, cropping and hashing, the shortcut filter, the frozen ResNet-50 pass, and one local epoch over 8.69M trainable parameters — while the server only receives, averages and broadcasts. No image, note, box, or per-sample quantity crosses the boundary; one float32 client-round costs 59.5–60.5 MiB up in the matched-setting suite.

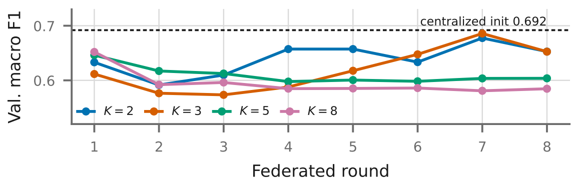

Fig. 7: Rounds against performance for the retained core, one line per client count. Dashed: the centralized initialization each run adapts from. K=2 and 3 climb toward it; K=5 and 8 are flat.

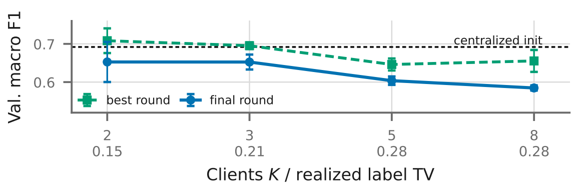

Fig. 8: Clients against performance for the retained core: the same fixed training set split over more owners. Bars are ±1 SD over 3 seeds; realized label TV is printed under each tick.

Source-group bootstrap intervals (5,000 replicates) resample source photographs rather than crops.

Five evidence tracks remain separate: the exact-pair same-checkpoint ablation; the validation-selected family comparison on the common 75-crop support; the 74-crop validation-only FedAvg sweep; earlier text, vision, and proxy-fusion screens whose differing supports permit only within-panel ranking; and the 371-pair matched-setting suite on genuine box-linked observations, with no label-matched resampling and no synthetic images.

B. Federated Client and Round Scaling

Figs. 7 and 8 answer the two federated scaling questions directly. Final-round macro F1 falls from 0.6527 at $K = 2$ to 0.5846 at $K = 8$ against a centralized initialization of 0.6919, so 84–94% is retained: adaptation costs something on the corrected corpus where it cost almost nothing on the saturated one. Round-wise means stay between 0.5735 and 0.6857, the highest a $K = 3$ transient at round 7 rather than the final model. Realized label-distribution total variation rises with K from 0.151 to 0.281 without a monotone final-score penalty, over 3 partition seeds.

This result is narrow: every run begins from the same centrally selected checkpoint, so it supports federated adaptation of the multimodal core, not superiority over training from scratch. Three partition seeds measure ownership-split

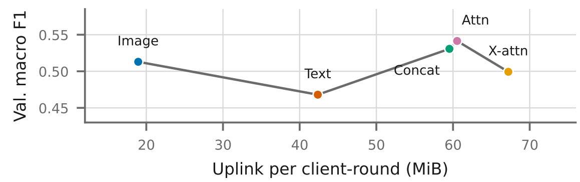

Fig. 9: Uplink cost against accuracy at α=0.1; the line orders architectures by uplink rather than interpolating. Cost spans 3.5×, accuracy 0.073.

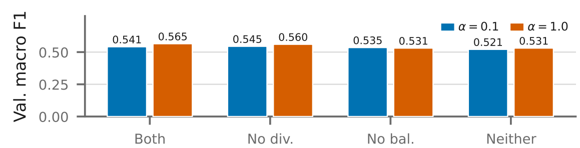

Fig. 10: Anti-collapse arms at both skews, from zero because the finding is that none of them collapses. Values are printed: the arms span 0.044, all of it the balanced sampler, and the diversity term does nothing.

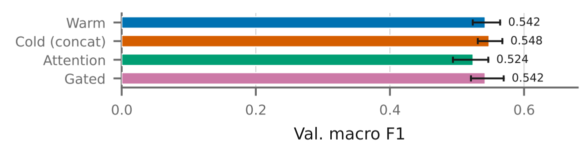

Fig. 11: Every single-architecture matched-setting arm: bars are 3-seed means, whiskers the observed seed range. The cold and concat arms are the same runs. The ranges overlap, so no arm clears the spread.

variability, not checkpoint-training variance, and simulated clients cannot establish estate-level privacy.

C. Matched-Setting Controlled Ablations

Table II closes the remaining gaps without mixing supports. The note correction rewrote text only, so the corrected splitter still realizes label TV 0.451 against 0.265 at $\alpha = 0.1$. Federation beats local-only training for all five architectures (+0.124 to +0.229), and the ordering is architectural rather than the note-leakage ceiling the previous corpus imposed. No aggregator dominates — FedProx leads on both fusion variants, FedAvg on image-only and cross-attention, FedBN on text — while SCAFFOLD is unstable everywhere (0.130–0.294). Communication spans 3.5×, 18.9 to 67.2 MiB per client-round, so image-only reaches 0.513 macro F1 at roughly a third the uplink of either fusion (Fig. 9). Two arms report nothing, shown rather than asserted: the anti-collapse compo-

TABLE II: Federated results on 371 genuine box-linked crop–note observations (260/55/56 over 199 independent source groups; no synthetic images). Every architecture within a cell trains on the identical client partition, so differences are architectural rather than partition effects. Values are 3-seed means of validation macro F1 over 50 federated rounds. Nothing saturates: panel (a) spans 0.431–0.561.

(a) FedAvg by architecture and skew

Architecture	α=.1	α=.5	α=1	α=10
Image only	0.513	0.516	0.479	0.486
Text only	0.468	0.448	0.441	0.431
Concat VLM	0.531	0.551	0.548	0.546
Attention VLM	0.541	0.555	0.524	0.561
Cross-attn. VLM	0.499	0.485	0.466	0.512

(b) Aggregator at α=0.1

Arch.	Avg	Prox	SCAF	BN
Image	**.513**	.504	.222	.196
Text	.468	.467	.220	**.470**
Concat	.531	**.560**	.179	.427
Attention	.541	**.549**	.130	.427
Cross-att	**.499**	.482	.294	.455

(c) Cost and federation gain

Architecture	MiB	Δ local
Image only	18.9	+0.215
Text only	42.4	+0.124
Concat VLM	59.5	+0.229
Attention VLM	60.5	+0.219
Cross-attn. VLM	67.2	+0.138

TABLE III: Federated adaptation across the paper's systems, not only ResNet+compact. Frozen encoders with SGD-trained heads on the shared source-grouped split; within a cell every system trains on the identical client partition. Entries are means of 3 seeds over 50 rounds and all client counts, so each is an average of 18 runs; *Local* is the no-communication control. Fusion leads under strong label skew, while the text branch leads at mild skew — the ordering depends on the skew and is reported, not ranked.

	$\alpha = 0.1$		$\alpha = 1$	
System	FedAvg	Local	FedAvg	Local
Text (frozen DistilBERT)	0.607	0.266	0.628	0.298
Image (frozen ViT-tiny)	0.508	0.323	0.559	0.391
ViT+DistilBERT fusion	**0.723**	0.317	**0.775**	0.364
ResNet+compact (warm)	0.892–0.910 over 2–8 clients			

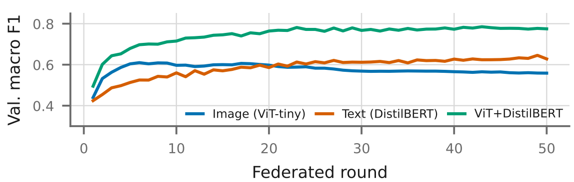

Fig. 12: Rounds against performance for the frozen-encoder systems at α=1 (three seeds × six client counts). Fusion leads at every round: 0.775 against 0.628 text and 0.559 image.

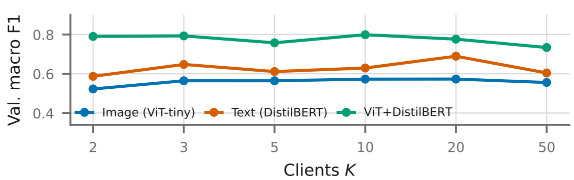

Fig. 13: The same systems against client count. Fusion leads at every count, 0.734–0.799 against 0.587–0.689 for text.

nents move the mean by at most 0.044 (Fig. 10), and warm start leads on two of three seeds but loses the mean by 0.006, with the fusion variants inside the same seed spread (Fig. 11). Neither is claimed necessary.

Table III extends the federated evidence beyond one architecture. A closed-form ridge head has no gradients for FedAvg to average, so these systems are federated as linear probes — frozen encoder, SGD head — which is not the centralized estimator. Within this protocol fusion leads at both skews (0.723 against 0.607 for text and 0.508 for image at $\alpha = 0.1$; 0.775, 0.628 and 0.559 at $\alpha = 1$) and takes the largest federation gain over local-only training in both cells (+0.406 and +0.411), and it holds at every round (Fig. 12) and every client count: at $\alpha = 1$ fusion spans 0.734–0.799 from $K = 2$ to 50, above text (0.587–0.689) throughout (Fig. 13). Combining the branches therefore helps most exactly where partitioning is most severe. Client partitions remain simulated.

D. Same-Checkpoint Ablation and Image-Only Gains

Both branches carry evidence and neither dominates: unimodally the image leads this test 0.6400 against 0.2400, and Section V-H quantifies both ablations. Notes are templated, so every note-side number is an upper bound, and the routing family was refined on this partition. A validation-selected test-time-augmentation search over 25 view/scale candidates reaches 74.32% validation accuracy (macro F1 0.6854) with the base, tight-crop and context-crop views, and 64.00% on the locked test (macro F1 0.5532). This is an inference-time ensemble over one backbone, not a stronger encoder; rare-class behaviour remains unstable.

E. Unimodal and Multimodal Systems, Centralized and Federated

Tables IV and V place a classical feature baseline, five frozen text encoders, five frozen vision encoders and their fusions on the same 222/74/75 support, scored centrally and again under FedAvg.

Unimodal. With complete notes the image branch leads the text branch: ViT-Base reaches 0.733 against 0.640 for the best text encoder, BERT-small. Federating costs little on either side (0.733 to 0.667; 0.640 to 0.582), so the modality separates these systems, not the deployment mode. The classical rows locate where pretraining earns its place: decisive on images (0.440 against 0.733) and absent on text, where TF-IDF with a linear SVM ties the best transformer at 0.640. Neither branch is near the ceiling, so the comparison is informative rather than saturated.

Multimodal. Fusion beats both parents outright, and does so at complete notes: ViT-Base with BERT-small reaches 0.800 against 0.733 for the best image-only encoder and 0.640 for the best text-only encoder. All five ViT-Base+BERT rows clear the best unimodal system (lowest 0.747 against 0.733), so the gain is not one lucky pairing; the classical TF-IDF with colour/HOG fusion does not, placing the gain in the pretrained encoders rather than in concatenation. The advantage widens as notes degrade: at 75% deletion text falls to 0.373 while fusion holds 0.787 and the image branch is unmoved by construction. Federation preserves the ordering rather than creating it, and its cost falls as the note thins — 0.800 to 0.667 at complete notes, 0.787 to 0.716 at 75% deletion.

Every ViT-Base pairing stays above the best text-only system at every sparsity, but which partner is best does not: BERT-small leads at complete notes, BERT-mini at 75%, and the spread across the five widens from 0.053 to 0.187. The evidence supports fusion over its parents, not a choice of text encoder.

F. Robustness: Skew, Dropout, Stragglers and Poisoning

Skew and dropout break nothing (Fig. 14a)**.** From α=10 down to α=0.01, and under a pathological partition giving each client a single class, fused macro-F1 stays in 0.741–0.776 — its worst setting above the best setting of either

TABLE IV: Unimodal systems on the common 75-crop test support, centralized and federated. Every row uses the identical 222/74/75 source-grouped split, the identical crop-linked notes, and the identical train-fitted leakage mask; configurations are chosen on the 74 validation crops and test is scored once. Federated rows are FedAvg over $K{=}5$ Dirichlet($\alpha{=}1$) clients for 50 rounds, averaged over 3 seeds, with the round chosen on validation only. Centralized heads are closed-form and federated heads are SGD-trained, because a closed-form solve has no gradients to average, so the two columns are close but not the same estimator. Image rows do not move with note sparsity by construction. This table varies the encoder under one fixed pipeline — frozen features with a single closed-form head — whereas Figs. 4 and 3 vary the pipeline for a fixed encoder. Bold marks the best value in each block and column.

		Complete notes				50% of notes deleted				75% of notes deleted			
		Centralized		Federated		Centralized		Federated		Centralized		Federated	
System	Dim.	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1
Text only													
TF-IDF + linear SVM	864	**0.640**	0.503	**0.582**	**0.469**	**0.493**	**0.390**	**0.476**	**0.375**	**0.373**	**0.298**	**0.360**	**0.293**
BERT-tiny	128	0.613	0.508	0.440	0.332	0.400	0.316	0.324	0.251	0.320	0.204	0.276	0.196
BERT-mini	256	0.613	0.496	0.427	0.331	0.427	0.318	0.440	0.332	0.280	0.186	0.276	0.190
BERT-small	512	**0.640**	**0.512**	0.564	0.467	0.427	0.306	0.431	0.331	**0.373**	0.291	0.253	0.197
BERT-medium	512	0.600	0.493	0.529	0.426	0.333	0.254	0.400	0.309	0.253	0.158	0.213	0.185
DistilBERT	768	0.613	0.485	0.529	0.422	0.427	0.328	0.431	0.332	0.360	0.286	0.249	0.194
Image only													
Colour+HOG SVM	630	0.440	0.346	0.471	0.392	0.440	0.346	0.471	0.392	0.440	0.346	0.471	0.392
Swin-tiny	768	0.600	0.507	0.569	0.463	0.600	0.507	0.569	0.463	0.600	0.507	0.569	0.463
ConvNeXT-tiny	768	0.560	0.493	0.560	0.470	0.560	0.493	0.560	0.470	0.560	0.493	0.560	0.470
EfficientNet-B0	1280	0.680	0.561	**0.680**	0.573	0.680	0.561	**0.680**	0.573	0.680	0.561	**0.680**	0.573
DeiT-tiny	192	0.693	0.545	0.636	0.520	0.693	0.545	0.636	0.520	0.693	0.545	0.636	0.520
ViT-Base/16	768	**0.733**	**0.615**	0.667	**0.577**	**0.733**	**0.615**	0.667	**0.577**	**0.733**	**0.615**	0.667	**0.577**

TABLE V: Multimodal systems on the same 75 crops, under the same protocol as Table IV. The best text-only and best image-only rows are repeated at the foot for reference. Fusion leads both parents at complete notes rather than only under note loss: the best fused system reaches 0.800 against 0.733 image-only and 0.640 text-only, and all 5 ViT-Base rows clear the best unimodal system. The margin over text then widens as notes are deleted; the image parent is flat by construction and is only matched at 50% deletion. Bold marks the best value in each column.

		Complete notes				50% of notes deleted				75% of notes deleted			
		Centralized		Federated		Centralized		Federated		Centralized		Federated	
System	Dim.	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1
Multimodal (image + text)													
TF-IDF + Colour/HOG SVM	1494	0.667	0.537	0.631	0.526	0.587	0.480	0.547	0.451	0.547	0.451	0.449	0.381
ViT-Base + BERT-tiny	896	0.747	0.625	**0.693**	0.597	**0.733**	**0.614**	**0.684**	**0.645**	0.600	0.513	0.698	0.610
ViT-Base + BERT-mini	1024	0.773	0.643	0.676	0.585	0.667	0.574	**0.684**	0.619	**0.787**	**0.655**	**0.716**	**0.614**
ViT-Base + BERT-small	1280	**0.800**	**0.664**	0.667	**0.600**	0.627	0.543	0.631	0.545	0.613	0.528	0.676	0.572
ViT-Base + BERT-medium	1280	0.773	0.642	0.676	0.566	0.720	0.593	0.653	0.561	0.680	0.571	0.622	0.515
ViT-Base + DistilBERT	1536	0.773	0.645	0.653	0.553	0.693	0.579	0.636	0.554	0.693	0.581	0.640	0.531
Best unimodal system, repeated for reference													
Text only: TF-IDF + linear SVM	864	0.640	0.503	0.582	0.469	0.493	0.390	0.476	0.375	0.373	0.298	0.360	0.293
Image only: ViT-Base/16	768	0.733	0.615	0.667	0.577	0.733	0.615	0.667	0.577	0.733	0.615	0.667	0.577

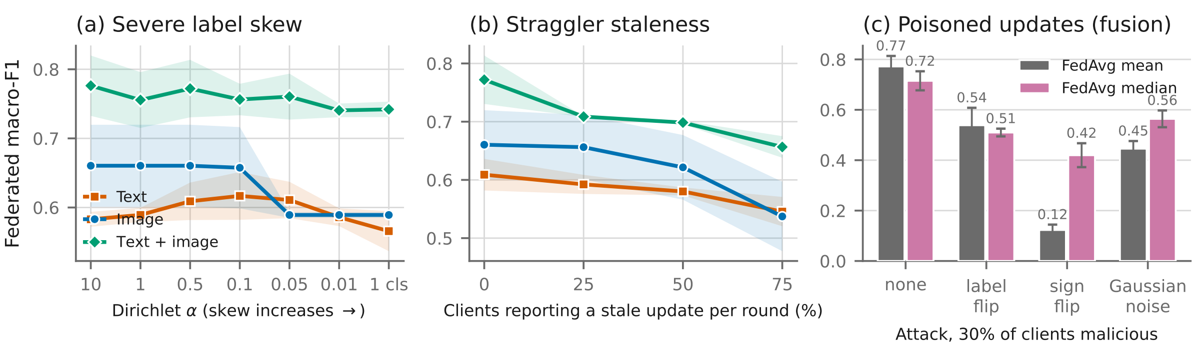

Fig. 14: Federated failure modes on the audited split. (a) Label skew from $\alpha{=}10$ to one class per client. (b) Clients reporting an update computed from an earlier global model. (c) Malicious clients under mean against coordinate-wise-median aggregation. Bands are ±1 SD over 3 seeds. Client dropout is omitted because it is flat to 70%; the released figure set carries it.

parent (0.617 text, 0.660 image). Dropout is flatter still: 70% of clients failing to report each round leaves the fused system at 0.773 against 0.772 with none. The image branch is the limiting case: a convex probe returns per-seed-identical scores across $\alpha{=}0.5$, 1 and 10 and again across $\alpha{=}0.05$, 0.01 and the pathological split, so skew changes the path and not the destination.

Stragglers cost what dropout does not (Fig. 14b). A stale

TABLE VI: Common-support encoder selection. Architecture choice uses validation only; test metrics are reported after selection. All rows are scored under the deterministic sparse-note mask at 15% of note tokens retained, the rate the routing hierarchy selects. The image row uses ViT-tiny, not the ViT-Base of Table IV.

Actual system	Val. acc.	Test acc.	Test F1
Text PLM (DistilBERT)	0.2297	0.2400	0.2167
Image ViT (ViT-tiny)	0.7432	**0.6400**	**0.5532**
Raw ResNet+compact VLM	0.7568	0.5867	0.5076
ViT+DistilBERT VLM	0.7297	**0.6400**	0.5334
Routed ResNet+compact (LEAF)	**0.7703**	0.6133	0.5319

update arrives and is averaged in, pulling the model toward an older global state rather than merely shrinking the cohort. At 75% stale updates per round the fused system loses 0.115 (0.772 to 0.657) and the image branch 0.123 (0.660 to 0.537), where the same fraction of *dropped* clients costs nothing. Discarding a late update beats waiting for it.

Poisoning is where plain FedAvg has no defence (Fig. 14c). Sign-flipped updates are catastrophic at every fraction: 10% malicious clients take the fused system from 0.772 to 0.298 and 30% to 0.123, with text (0.136) and image (0.104) equally destroyed — the one condition in which fusion is not best, because all three are broken. Label flip degrades gradually instead (0.658, 0.611, 0.539 at 10/20/30%) and Gaussian noise sits between them. Coordinate-wise median recovers the outlier attacks (sign flip to 0.613, 0.543, 0.420) but not label flip, which is worse at 30% (0.510 against 0.539): a label-flipped update is a well-formed gradient step toward the wrong target, not an outlier in parameter space. The median also costs 0.057 with nobody attacking, so it is a trade, not a free defence. Nothing here is Byzantine-tolerant.

G. Actual Model-Family and Inter-Model Comparison

In Table VI the routed ResNet–compact model, LEAF, is the *incumbent* — the core already carried into the routing and federated experiments — and the pooled ViT–DistilBERT model is the *candidate*, adopted only if it beats the incumbent on validation. Validation selects the incumbent before test extraction: 77.03% accuracy and 0.7169 F1 against 72.97% and 0.5980 for the candidate. On test the candidate predicts 2 more crops, but the 11-versus-9 discordance is not significant (McNemar $p = 0.82$) and the two macro-F1 values differ by 0.0014. On 75 crops the support does not separate them, so the selection rule decides and the test column is reported rather than ranked. The enhanced gate gains eight correct crops over the direct image path and the router 15, mainly in blight and looper; two leaf-hopper crops are too few to carry a claim.

H. Missing-Modality and Corruption Probes

Mismatching the note costs only 2.67 accuracy points and 0.0187 F1, and exact cross-modal recall@1 is 1.33% against class-level 0.600/0.613: the embedding is organized by class, not by crop identity.

The mismatch result shows dependence on the note, not whether the image is used at all. We evaluate the retained checkpoint on the locked 75-crop test under graded corruption of each modality and two full image ablations, comparing predictions crop-by-crop against the clean run.

Both branches are load-bearing. Table VII reports the sweep. Deleting every attended text token costs 0.226 macro F1 and moves 41 of 75 predictions; replacing the image with uniform noise costs 0.429 and moves 61. The image now carries more of the decision than the note, reversing the previous corpus, but neither branch is inert. The reliability gate does not track this: mean w_t is 0.263 on clean input and rises to 0.361 under complete text deletion and 0.380 under complete image noise, so it weights a destroyed modality more rather than less.

TABLE VII: Modality corruption on the locked 75-crop test, retained checkpoint. Flips count predictions differing from the clean run. Neither modality is inert and neither dominates; uniform noise of matched statistics is the most damaging condition tested, which separates robustness to sensor noise from dependence on image content.

Condition	Acc.	Macro F1	Flips
Clean	0.6267	0.5163	—
Text deletion 25%	0.6133	0.5060	4/75
Text deletion 50%	0.6267	0.5244	7/75
Text deletion 75%	0.6000	0.5068	11/75
Text deletion 100%	0.3600	0.2903	41/75
Image Gaussian σ=0.09	0.6400	0.5234	13/75
Image Gaussian σ=0.18	0.6400	0.5308	16/75
Image Gaussian σ=0.26	0.6267	0.5247	17/75
Image Gaussian σ=0.35	0.5867	0.4906	25/75
Image zeroed	0.3733	0.1603	49/75
Image uniform noise	**0.2800**	0.0875	**61/75**

Ablation strength determines the verdict. Additive Gaussian noise at σ=0.35 leaves leaf and lesion plainly visible and moves 25 of 75 predictions, whereas zeroing the image moves 49 and replacing it with uniform noise moves 61. An earlier 30-pair proxy screen used the Gaussian setting alone and read the visual branch as inert; that reading does not survive genuine removal. Evidence that a branch is unused requires an ablation that removes it, not one that degrades it.

I. Architecture Component Ablation

Eleven variants $\times$ 3 seeds on one device (33 runs) separate component effects from seed noise, which the earlier CPU-versus-H100 comparison confounded. The floor is high: pooled within-variant SD is 0.032 accuracy and the full model alone spans 0.053 across seeds (0.649 mean). One effect clears it — a lightweight vision encoder costs 0.227 — and no other ablation moves the mean by more than 0.071. The expert-residual weight is unsupported: λ=0 (-0.013) beats λ=1 (-0.044) and λ=4 (-0.040) against the λ=2 default, so the ordering is non-monotone. With a frozen-encoder concatenation reaching 0.800 on the same support (Table V), the trained architecture is not what produces the multimodal result and is not claimed to be.

J. Advisory Retrieval Results

Retrieval reaches 90.05% correct-class precision@1 against a 20.01% chance rate, rising to 95.52% with the classify stage. That second number *is* the classifier's accuracy, because retrieval restricted to one predicted class can only return that class, so routing rather than ranking decides correctness. A stratified query audit locates the failure: open retrieval is near-saturated (99.24%) on the queries carrying a class-unique sentence and falls to 72.46% on the rest, which routing lifts to 89.86%. Figure 15 shows where the misses go: looper caterpillars is the weak row, losing queries to blight, rust and mosquito bug, whose damage vocabulary it shares.

Three limits apply. The corpus is template-composed, so base and queries share sentence vocabulary even though no observation is shared, and these are upper bounds relative to free-form notes. The fusion re-ranker, the OBB overlay and the agronomic correctness of the advice remain unevaluated: this is a retrieval result, not evidence that the treatment is right.

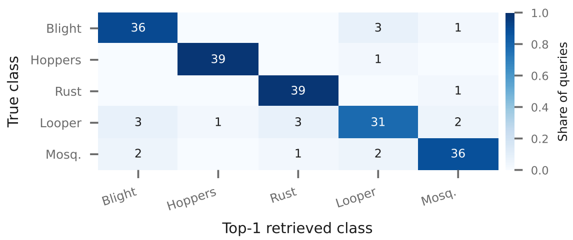

Fig. 15: Advisory retrieval confusion: true class against top-1 retrieved class, row-normalized so the diagonal is per-class precision@1; cells are query counts. Looper caterpillars is the weak row (31/40); hoppers and rust retrieve almost perfectly.

K. Reproducibility and Threats to Validity

The released bundle records data identity, checkpoint state, every federated round and each realized client partition; deterministic grouping, strict reload and validation-only adaptation are reproducible, but external estate transfer is not inferred from them. The locked test is small (38 sources, 75 crops, two leaf-hopper crops), and selection on 74 validation crops can overfit. The central checkpoint has one seed; FL seeds vary only client partitions, and clients are simulated. DistilBERT is a PLM rather than a generative LLM. The corruption probe is one seeded realization per severity and carries no interval. The robustness sweep federates linear probes on frozen features, so its invariances belong to that estimator and not to the deep model. Deployment requires multi-estate data, independent notes, rare-class coverage, and repeated end-to-end training.

VI. Conclusion

On 371 linked crops, validation retains LEAF, the ResNet–compact-Transformer router; the ViT–DistilBERT candidate's 2-crop test edge is non-significant ($p = 0.82$), and a 33-run multi-seed sweep leaves only vision-encoder capacity clearing the seed noise floor. FedAvg retains 84–94% of the 0.6919 initialization over 2–8 simulated clients. Fused federated training absorbs severe label skew and client dropout (0.741–0.776 macro F1 down to one class per client, unchanged at 70% dropout) but not staleness or poisoning: 30% sign-flipping clients take it from 0.772 to 0.123, and coordinate-wise median recovers only part of that. In the matched suite, FedAvg averages 0.510/0.492 across the five architectures at $\alpha = 0.1/1$, warm start no longer helps, and updates cost 18.9–67.2 MiB. No E3 arm collapses, so the stack is not claimed necessary. Multi-estate, co-observed evaluation remains required.

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